

Deep Semantic Plagiarism Detection Workspace

Abstract

Traditional plagiarism detection systems primarily rely on lexical string-matching algorithms, which analyze surface-level text overlaps, character sequences, and word-for-word duplication. While highly effective at catching raw copy-paste infractions, these lexical configurations fail significantly against advanced structural rewriting, synonym substitution, and AI-assisted paraphrasing—often referred to as "smart plagiarism." To address this vulnerability, this project introduces a Deep Semantic Plagiarism Detection Framework that shifts the tracking paradigm from syntax sequence matching to deep conceptual and contextual alignment.

The core system architecture leverages Transfer Learning via a pre-trained, state-of-the-art MPNet (Masked and Permuted Pre-training) Transformer model (all-mpnet-base-v2). The computational pipeline structures unstructured multi-page PDF documents by extracting native text via a PyPDF engine, segmenting the text stream into individual logical sentence blocks using the natural language tokenization rules of NLTK, and feeding these fragments into the deep neural network.

The model maps each tokenized sentence into a dense, high-dimensional vector space (768 dimensions), where clauses are geometrically embedded based on their underlying semantic properties rather than literal string composition. To evaluate the conceptual proximity between the student submissions and reference papers, a pairwise spatial alignment matrix is calculated dynamically using a vectorized Cosine Similarity metric. Sentences crossing a user-defined AI Alignment Sensitivity Threshold are flagged as plagiarized in real time.

To bridge the gap between complex deep learning outputs and end-user utility, the system is deployed as an interactive web client dashboard built with Gradio. Experimental analytics indicate that the framework achieves high stability and semantic sensitivity, exposing deliberate paraphrasing that completely bypasses classic lexical trackers. This workspace successfully demonstrates a robust, scalable natural language processing (NLP) application engineered for modern academic integrity verification.

Name: B. Sathwik Chaithya
Roll Number: 25011D0501

Dr. G. Vijaya Kumari
Professor